MADRAS INSTITUTE OF TECHNOLOGY

ANNA UNIVERSITY

AD23403-DISTRIBUTED SYSTEMS AND CLOUD COMPUTING

PROJECT REPORT

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Drone Command Center for Real-Time Monitoring and Mission Assignment

INTRODUCTION

The Drone Command Center (DCC) is a real-time system developed to monitor and control multiple drones efficiently. It collects and processes telemetry data such as location, battery level, temperature, and signal strength. The system uses a Spring Boot backend with a PostgreSQL database to store and manage data, along with an interactive web interface for visualization. It enables users to track drone movements on a live map, detect high-risk conditions, and assign missions dynamically. This ensures improved safety, better decision-making, and efficient drone management.

Objectives

- To monitor multiple drones in real time
- To track drone location and movement on a map
- To store telemetry data (battery, temperature, signal, etc.)
- To detect high-risk drone conditions
- To assign missions dynamically to available drones
- To provide an interactive and user-friendly dashboard

Technology Stack

- **Frontend:** HTML, CSS, JavaScript, Leaflet.js
- **Backend:** Spring Boot (Java)
- **Database:** PostgreSQL (Render Cloud)
- **ORM Tool:** Spring Data JPA
- **Web Service / Deployment:** Railway
- **API Communication:** REST APIs (JSON)
- **Additional Features:** Drone Simulation, Geocoding API, Real-time Polling

WORK FLOW OF THE PROJECT

Step 1: Drone Initialization and Simulation
The system initializes multiple drones with unique IDs, initial positions, and default parameters such as battery level, temperature, and signal strength. Each drone continuously updates its data through simulation, including movement, battery consumption, and environmental changes.

Step 2: Data Processing in Backend
The generated drone telemetry data is sent to the Spring Boot backend. The backend processes this data and converts it into structured objects (DroneRecord) using business logic implemented in the service layer.

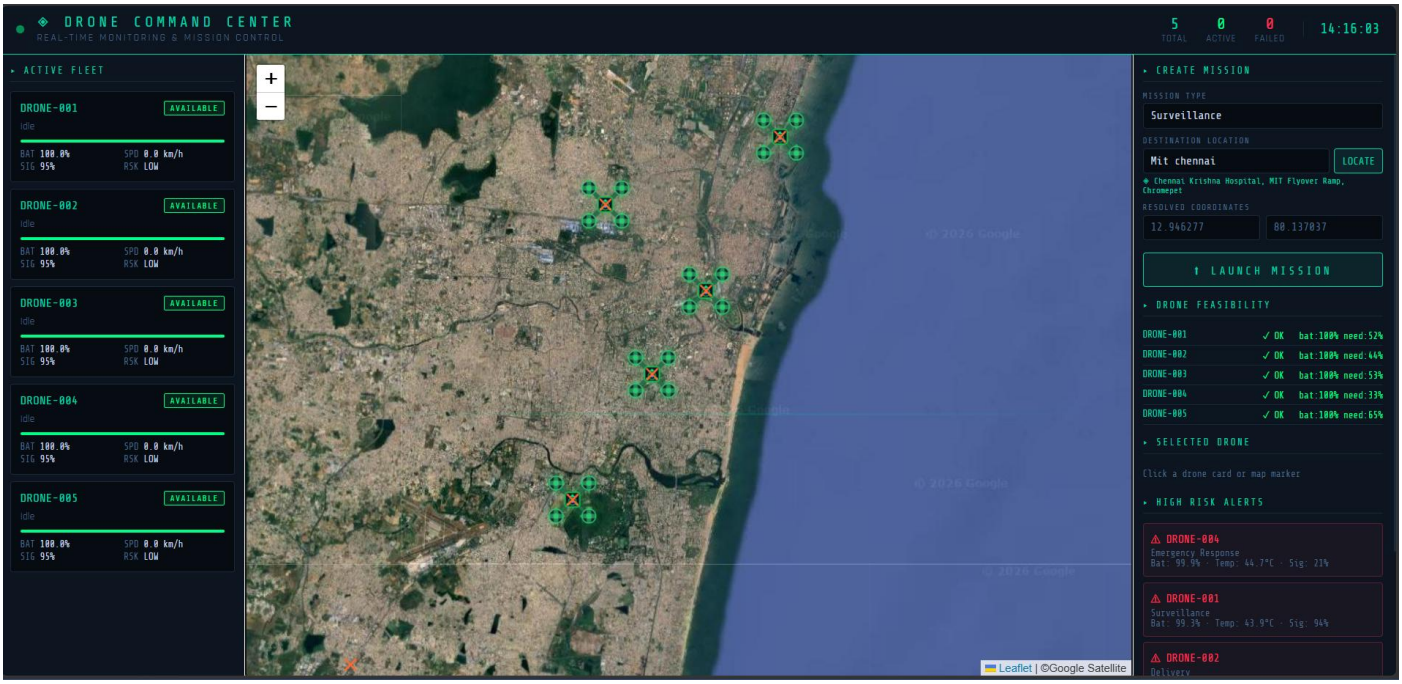
Step 3: Data Storage in Database
The processed data is stored in the PostgreSQL database. Each record contains detailed information such as drone location, battery level, temperature, status, and risk score, allowing both real-time and historical data analysis.

Step 4: API Communication and Data Fetching
The backend provides REST APIs that are periodically called by the frontend. The frontend fetches updated drone data every few seconds in JSON format, ensuring real-time synchronization between the system components.

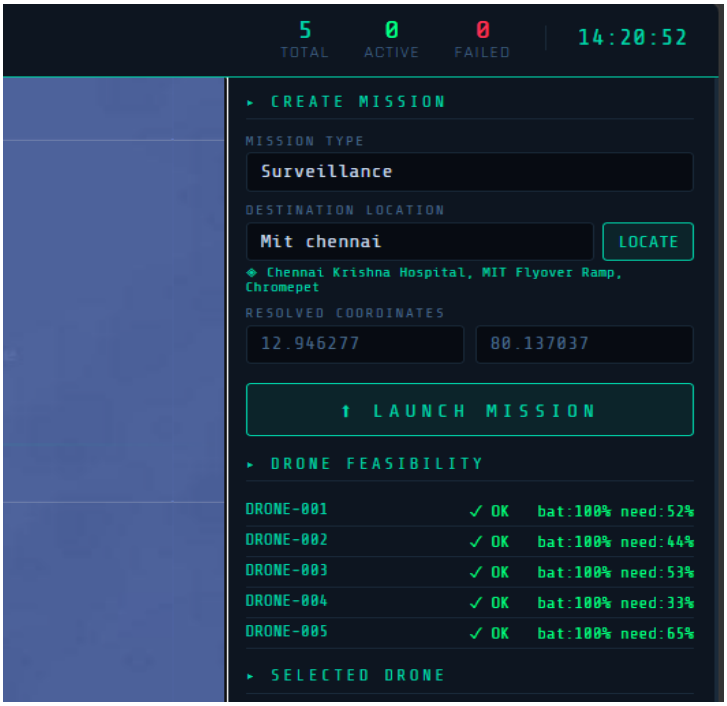
Step 5: Visualization and Mission Control
The frontend displays drone data on an interactive map using Leaflet.js and shows detailed information in dashboard panels. Users can assign missions by selecting a task and destination, and the system checks feasibility before assigning the mission to a suitable drone.

Step 6: Monitoring, Alerts, and Mission Completion
The system continuously monitors drone conditions and detects high-risk scenarios such as low battery or weak signal. Alerts are displayed on the dashboard for quick action. Once a drone reaches its destination, the mission is completed, and the system updates the drone status while displaying mission summary details.

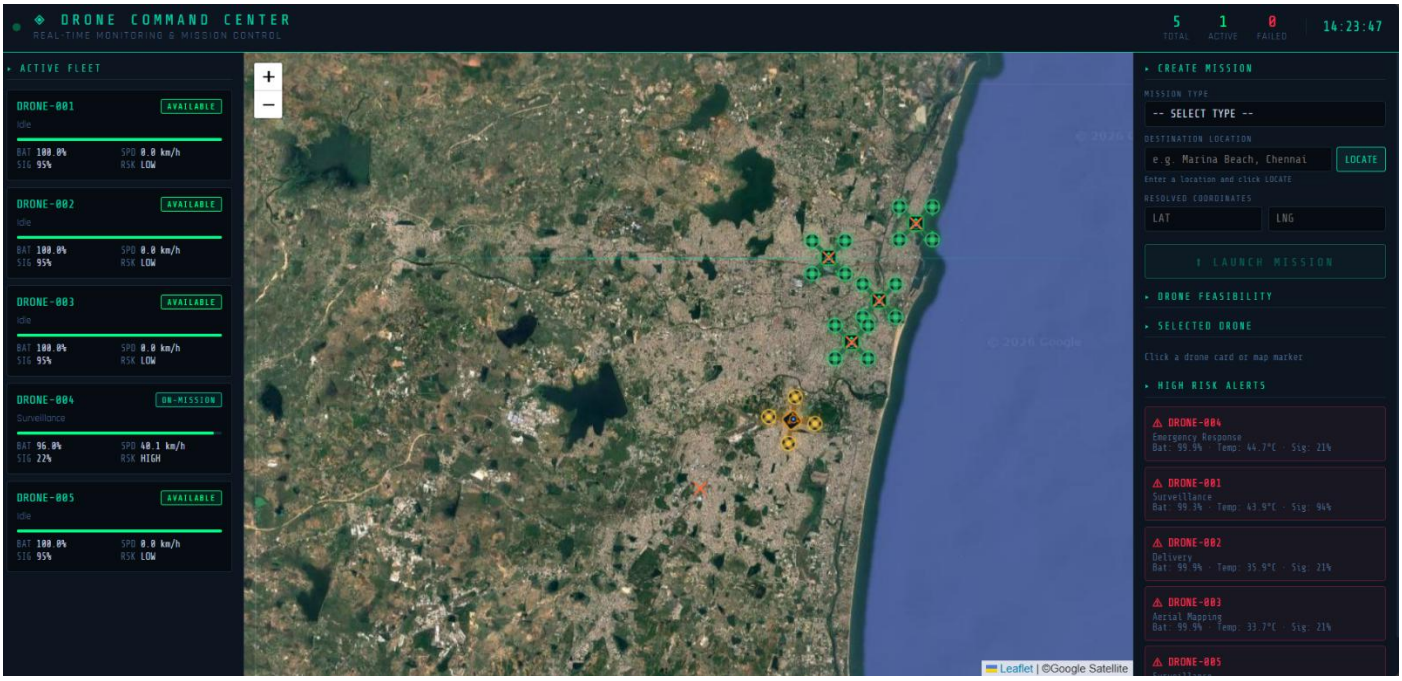
PICS OF THE PROJECT:



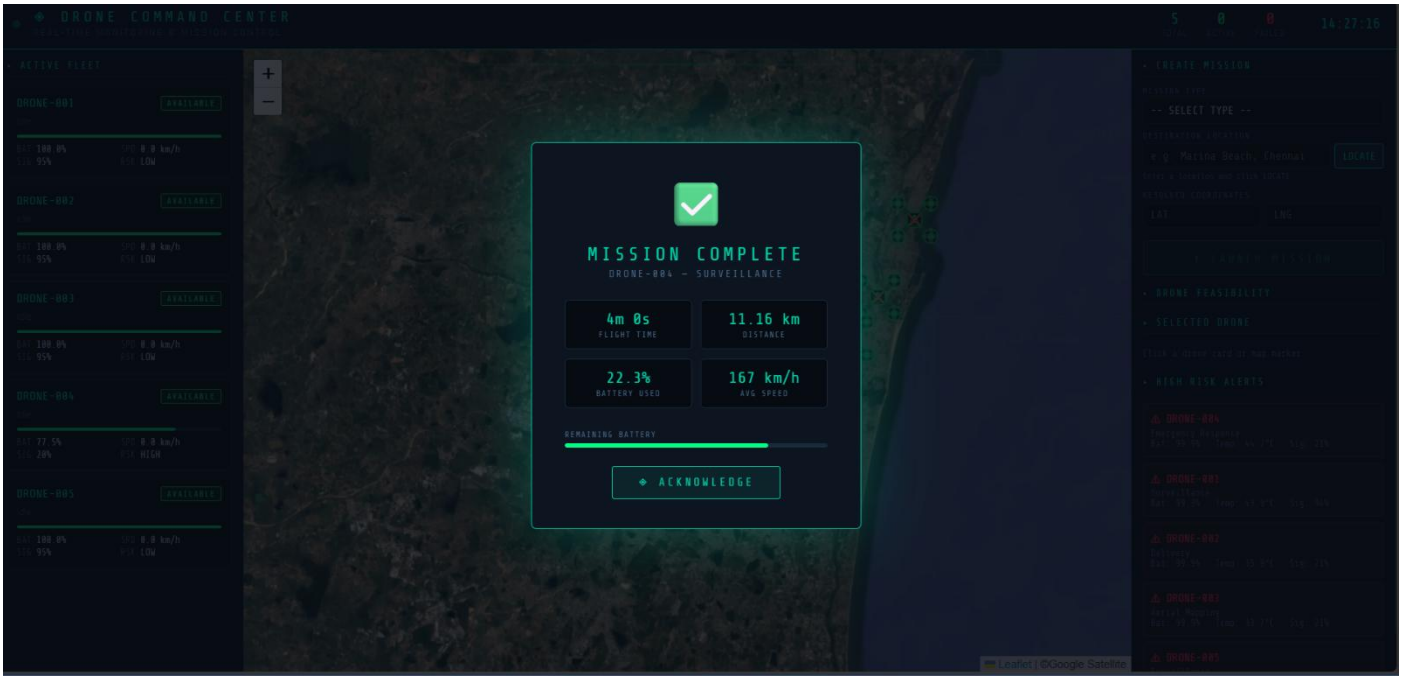
The above figure illustrates the initial setup of the Drone Command Center system, where a satellite map interface displays five initialized drones at their respective locations. This represents the default state of the system before mission assignment and real-time operations begin.



In this step, the user provides input to assign a task to the drone by selecting the mission type (e.g., Surveillance) and entering a destination location. The system uses a geocoding process to convert the entered location (e.g., MIT Chennai) into precise latitude and longitude coordinates. These coordinates are automatically filled in the respective fields. Based on this input, the system also performs a feasibility check to determine whether each drone has sufficient battery to complete the mission. Once the inputs are validated, the user can initiate the mission by clicking the 'Launch Mission' button.



The above figure represents the system after a mission has been successfully assigned to a drone. One of the drones (e.g., DRONE-004) is now in the 'on-mission' state, and its movement towards the assigned destination is visualized on the satellite map. The drone's real-time parameters such as battery level, speed, signal strength, and risk status are continuously updated and displayed in the active fleet panel. The map shows the drone's current position along with its destination, and its path is dynamically tracked. Additionally, the system highlights high-risk conditions in the alerts panel, enabling quick monitoring and decision-making during the mission.



Once the drone successfully reaches its destination, the system marks the mission as completed and displays a summary of the mission details. This includes key performance metrics such as total flight time, distance traveled, battery consumed, and average speed. The remaining battery level is also visually represented. After completion, the drone status is updated back to 'available', making it ready for future missions. This feature provides a clear overview of mission performance and helps in analyzing the efficiency of drone operations.

Societal Impact

- Helps in **emergency response** during disasters (rescue, surveillance)
- Enables **fast and efficient delivery** in remote areas
- Improves **security and surveillance** in sensitive zones
- Supports **agriculture** through crop monitoring and analysis
- Assists in **mapping and surveying** for infrastructure planning
- Enhances **public safety** with real-time monitoring

Future Enhancements

- Integration with **real-time physical drones (IoT)**
- Implementation of **AI-based risk prediction systems**
- Adding **user authentication and security features**
- Development of a **mobile application** for remote access
- Advanced **map analytics (route optimization, heatmaps)**
- Cloud scaling to support **large-scale drone fleets**
- Real-time **live video streaming from drones**

Conclusion

The Drone Command Center (DCC) is a real-time system for monitoring and controlling drones. It uses a Spring Boot backend, PostgreSQL database, and an interactive frontend to track drones, manage missions, and detect risks. The system ensures efficient operation and demonstrates a practical full-stack solution for modern drone applications.